Module 01: Systems in Robotics

Learning Objectives

1. Define **Systems** within the context of Robotics.
2. Analyze how Systems interacts with other components of the Active Inference framework.
3. Apply specific constraints of Robotics to the formal definition of Systems.

Introduction

This module explores **Systems**. In the **Robotics** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. Systems is a critical component of the 8-part Active Inference spine, bridging the gap between Planning and Agents.

Key Concepts

1. Systems as a Markov Blanket Boundary

How does Systems define the boundary between the agent and the environment?

2. Generative Models of Systems

What parameters involved in Systems must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of Systems drive the perception-action loop?

Applications

In Robotics, we see Systems manifest in: * **Specific Example 1**: A Waymo-class self-driving vehicle defines its system boundary through a multi-modal sensor suite (LIDAR, cameras, radar, ultrasonics) and actuator interface (steering, throttle, brake) that together form the Markov blanket; the vehicle's internal states (path planner, prediction module, localization stack) are conditionally independent of the environment given these boundary elements, and the entire autonomous driving software stack can be understood as a single system performing hierarchical free energy minimization across perception, prediction, and motion planning subsystems. * **Specific Example 2**: A multi-robot search-and-rescue system composed of heterogeneous agents (ground rovers, aerial drones, and a stationary base station) forms a distributed system where each agent's individual Markov blanket is nested within the team-level system boundary; the base station aggregates partial maps from each agent's local SLAM system, and the team-level generative model of the disaster site emerges from the composition of individual agent models connected through shared spatial reference frames and communication links.

Conclusion

Understanding Systems allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.